

Grammar $\langle \Sigma, V, S, P \rangle$ rules $l \rightarrow r$

FSA = Reg

PDA $\nwarrow$ Context free
 PDA $\swarrow$

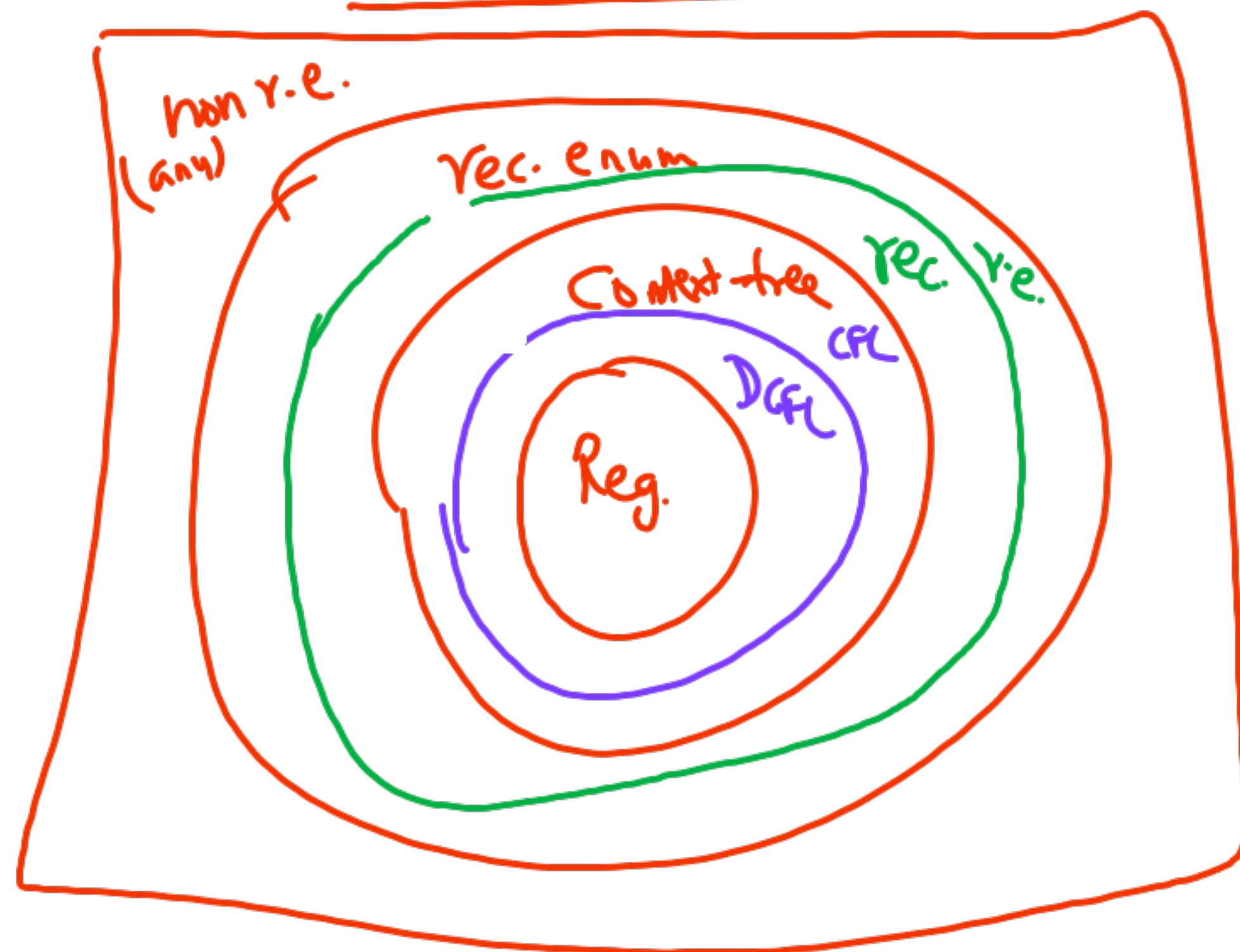
TM Type-0
 r.e.

$l \in V$ (simple var)	left-linear $l \rightarrow XaTc$ right-linear $l \rightarrow abcY$
$l \in V$ (simple var)	any string $(\Sigma \cup V)^*$
any	any

CSL $\nearrow$

reg-exp
 $(0+1)^* 10^*(\dots)$

CHOMSKY-Hierarchy



Exam.

Counting

Enumeration

λ -calc

Equational Prog.

Recursive Function Theory

Topics

Closure Properties

$L_1 \cup L_2, L_1 \cap L_2$
 $\vdots$
 $L_1 - L_2$

Decision Problems

Is $w \in L$?

Normal Forms

Chomsky /
Greibach

Undecidability

NP-completeness.

Post Cor Problem

Time Complexity
of

Decision Problems

Recursive

TM always answers Yes/No

$P \stackrel{?}{=} NP$

Verify guessed soln in polynomial # of steps

2nd tape

w $|w| = n$

DTM

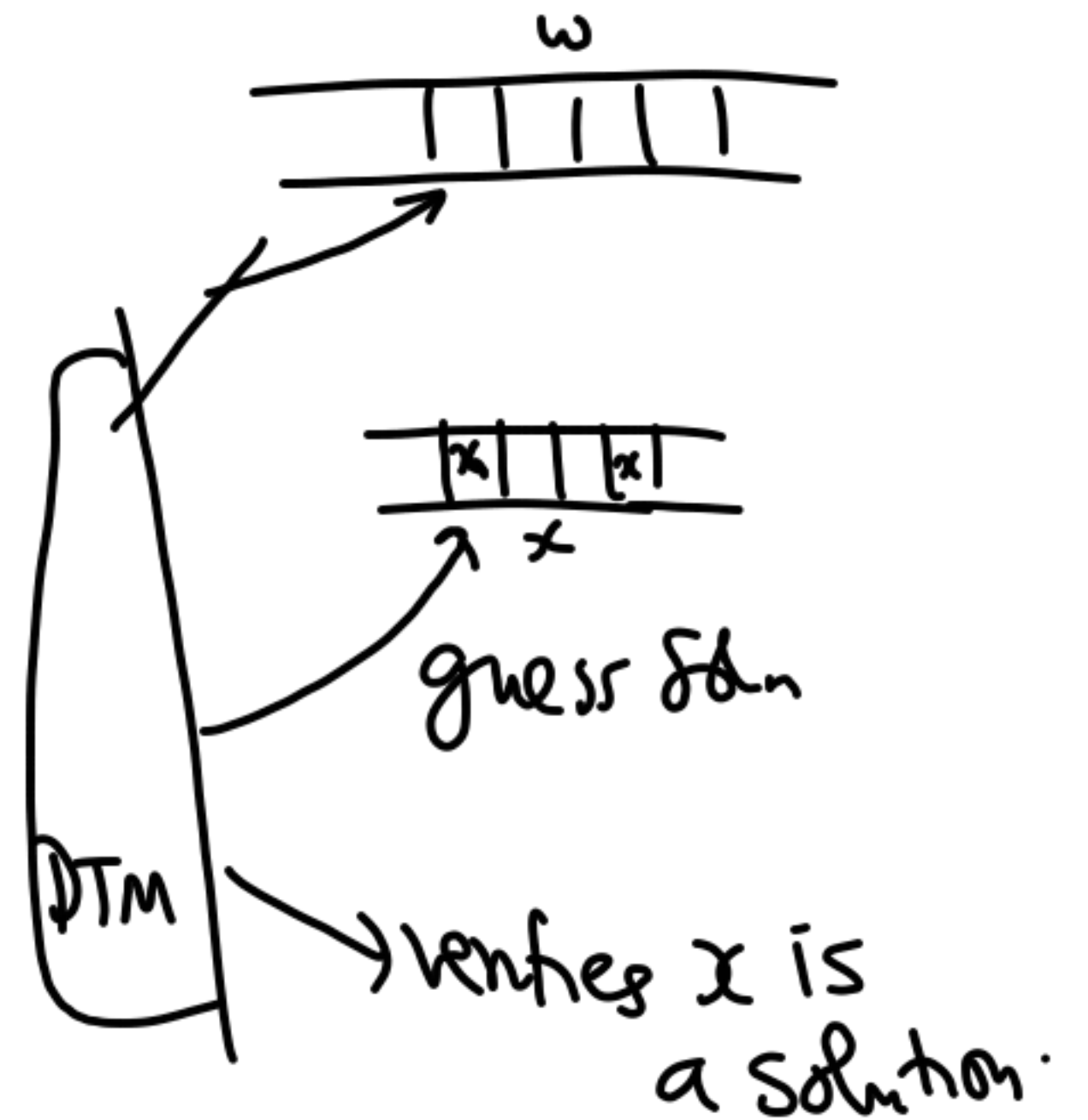
in polynomial number of steps

$\langle q_0, w \rangle \vdash \dots \vdash \text{Yes}$

$\vdash \vdash \vdash$ or No

$n^3 \quad n^8$

$7n^3 + 9n^2 + \dots$



$P \stackrel{?}{=} NP$ NP-complete

SAT

most difficult problem in NP.

Polynomial time reduction from any other problem

"Stanford" Lecture 2

Polynomial-reduction

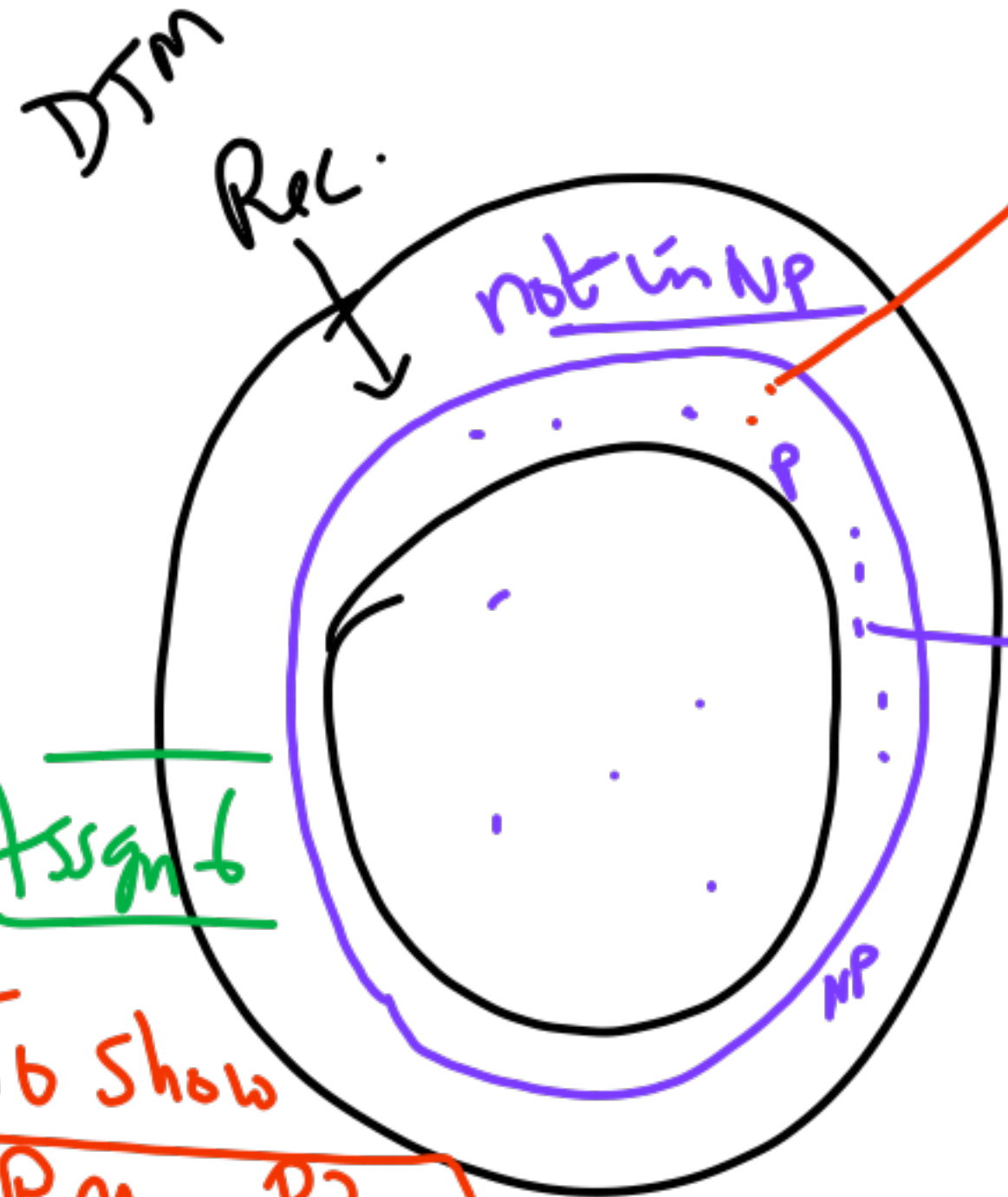
$SAT \leq_p 3-SAT$ — Vertex-Cover
Knapsack

Solution preserving

Known to be in NP
but not known to be in P

$P \subseteq NP$

Strict inclusion
reduces in poly-time to P3.



Assign 6

To show
Problem P3
is NP-complete

① Show $P3 \in NP$

② Show known NP-complete problem reduces in poly-time to P3.

